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Optimizing exercise interventions for motor skills enhancement in attention deficit hyperactivity disorder: Evidence from meta-analysis

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Abstract

Objective: This meta-analysis aimed to evaluate effects of physical exercises on gross and fine motor skills in children and adolescents with attention deficit hyperactivity disorder (ADHD), and to explore the influencing factors and dose–response relationships for optimizing exercise parameters.

Data sources: Six electronic databases were searched from inception to 20 June 2025, with an updated search performed on 27 January 2026.

Methods: The effect sizes were estimated using Hedge's *g* within three-level random-effects modeling frameworks. Subgroup analyses were performed to examine potential moderators, while restricted cubic splines were employed to characterize dose–response relationships.

Results: Twenty-one studies with 850 participants were included. Physical exercises showed moderate positive effects on both gross motor skills (Hedge's $g = 0.62$, 95% CI: 0.48–0.76) and fine motor skills (Hedge's $g = 0.72$, 95% CI: 0.54–0.90). Significant improvements were observed in subcomponents of gross motor skills (locomotion, object control, body coordination, and strength) and fine motor skills (manual dexterity, hand-eye coordination, and handwriting). Moderator analyses indicated that longer session durations (>45 min) and higher frequencies (≥3 sessions/week) were associated with greater benefits. Fine motor skills (~1350 min) appeared to require greater cumulative exercise exposure than gross motor skills (~810 min) to achieve substantial gains.

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Conclusion: Physical exercise is an effective intervention for improving motor skills in children and adolescents with ADHD. However, the effectiveness of interventions appears to be moderated by motor subcomponents and dose parameters. Tailored programs, specific motor skill targets and dose parameters, should be integrated into clinical and educational practices to maximize outcomes.

Keywords
Attention deficit hyperactivity disorder, physical exercises, children and adolescents, motor skills

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Introduction

Attention deficit hyperactivity disorder (ADHD) is a prevalent neurodevelopmental disorder characterized by inattention, hyperactivity, and impulsivity, affecting approximately 5.3%–7.2% of children worldwide.[1] Beyond these behavioral manifestations, up to 50% of children with ADHD exhibit impaired gross and fine motor skills, including difficulties in coordination, balance, object control, and manual dexterity, which often limit participation in physical activities, hinder social integration, and reduce quality of life.[2–9]

Pharmacological treatments are the most common intervention for ADHD but are associated with side effects (e.g., decreased appetite, sleep disturbances).[10] Nonpharmacological approaches, such as psychological and behavioral therapies, are often costly, less scalable, and have limited efficacy for motor deficits.[11] Physical exercises has emerged as a promising alternative, as it is easy to implement and has shown benefits for motor proficiency. For instance, studies have demonstrated table tennis and aquatic training can improve locomotor and object–control skills.[12–14] However, findings remain inconsistent, particularly regarding fine motor skills and handwriting.[12,15] These discrepancies likely stem from variations in exercise parameters (type, frequency, intensity, duration) and differences in the specific motor components assessed.

Recent meta-analyses have explored exercise effects on ADHD-related outcomes but leave critical gaps. Our previous work focused on symptoms and executive functions, with limited analysis of motor skills.[11,16] Prior syntheses either combined ADHD with other neurodevelopmental conditions, omitted fine motor subcomponents such as handwriting, or did not examine dose–response relationships or moderators (e.g., session duration, frequency).[15,17] Consequently, the field lacks a comprehensive, domain-specific evidence synthesis to guide intervention optimization.

Therefore, this study aimed to systematically analyze the effects of physical exercises on both gross and fine motor skills in children and adolescents with ADHD, with three key innovations: (1) disaggregating motor outcomes into specific subcomponents (e.g., locomotion, object control, manual dexterity, and handwriting) to identify which skills are most responsive to exercise interventions; (2) conducting moderator analyses to identify optimal intervention parameters (e.g., session duration, frequency); and (3) modeling dose–response relationships separately for gross and fine motor domains to inform intervention dosing. This study was essential for advancing the field of ADHD interventions and improving the quality of life for children and adolescents affected by this disorder.

Methods

This meta-analysis was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA).[18] Additionally, the protocol of this study has been registered in the International Prospective Register of Systematic Reviews (PROSPERO; ID: CRD420251064443).

The search strategy was formulated according to the patient Population, Intervention, Comparison, Outcome (PICO) framework. We searched the

following electronic databases, including PubMed, Web of Science, Cochrane Central Register of Controlled Trials, Embase, SPORTDiscus, and CNKI, for relevant studies. To avoid the risk of missing relevant studies, additional articles were searched manually in Google Scholar and bibliography of previous published relevant reviews. We used a combination of Medical Subject Heading terms with free text terms as follows: ADHD, attention deficit hyperactivity disorder, children, adolescents, physical activity, exercise, sport, motor, motor ability, movement skill, motor skill, gross motor, fine motor, motor development, motor proficiency. In each database, key words were used the Boolean operations "AND" and/or "OR" to combine connections. We searched all databases from their inception to 20 June 2025, and no language restrictions were applied. We conducted an updated search on 27 January 2026, and no additional studies meeting the inclusion criteria were identified in this update. The detailed search strategies are given in Supplementary Material Table S1.

Studies were included if they met the following criteria:

1. Participants: Children and adolescents aged 5–20 years who had received a formal diagnosis of ADHD based on standardized criteria, with no restrictions on gender or ADHD subtypes.
2. Interventions: Any structured physical exercise program lasting ≥3 consecutive weeks, with no restrictions on frequency, intensity, duration, or modality.
3. Comparator: Control conditions included passive control (e.g., no intervention, waiting-list) and active control (e.g., routine school activities).
4. Outcomes: The outcome measures included at least one objective and validated measure of motor proficiency/skills, including gross motor skills (e.g., locomotion, objective control, body coordination, balance, strength, flexibility) or fine motor skills (e.g., manual dexterity, hand-eye coordination, handwriting).
5. Study Design: Randomized controlled trials, quasi-randomized controlled trials, and nonrandomized comparison studies published in any language.

The exclusion criteria were as follows: (1) interventions combining physical exercises with additional therapeutic components (e.g., pharmacological treatment, cognitive-behavioral therapy, or dietary modifications) where the independent effect of exercise could not be determined; (2) single-session or acute exercise interventions; (3) noninterventional study designs, including protocols, reviews, cohort studies, case–control studies, conference papers, and book chapters; (4) studies lacking sufficient statistical data (e.g., means, standard deviations, or sample sizes) required for effect size calculation in meta-analysis.

The study selection process was conducted independently by two reviewers using predefined inclusion and exclusion criteria. Any discrepancies between the reviewers were resolved through discussion, and a third reviewer was consulted if consensus could not be reached. The consistency between reviewers during study selection was quantified using Kappa coefficient, computed with IBM SPSS Statistics 25.0 (SPSS Inc., Chicago, IL). Consistency levels were categorized as follows: excellent (>0.75), good (0.60–0.75), fair (0.40–0.59), or poor (<0.40).[19]

A standardized data extraction form was used to capture relevant data from each included study, including bibliographic information (authors, publication year), study design, participant characteristics (sample size, sex ratio, age), intervention details (type, frequency, duration, intensity, and length), comparator information, and outcome measures. Data extraction was performed independently by the two reviewers, and any discrepancies were resolved through discussion or consultation with a third reviewer. Data from each study were extracted and converted into a format suitable for meta-analysis. We extracted pre- and postintervention means, standard deviations, and sample sizes from all studies. The difference in means was calculated as[20]:

$$\text{Mean difference} = \text{Mean}_{\text{post}} - \text{Mean}_{\text{pre}}$$

For studies employing reverse-scored measures (where higher values represented poorer outcomes),

we inverted the scale by multiplying group means by −1.

The standard deviation of the difference (SD_{diff}) was calculated using the formula[20]:

$$SD_{diff} = \sqrt{SD_{pre}^2 + SD_{post}^2 - 2 \times 0.5 \times SD_{pre} \times SD_{post}}$$

When standard deviations were unavailable, we calculated them from standard errors, confidence intervals (CIs), t or p values.[20] For unreported data, we made at least three email attempts to contact corresponding authors. Graphical data were extracted using GetData Graph Digitizer (v2.20) when numerical results were only presented in figure format.

The Risk of Bias version 2 (RoB2) tool was employed to evaluate the quality of the included studies.[21] It comprehensively evaluated the RoB in the included studies from five key domains, namely the randomization process, deviations from the intended interventions, missing outcome data, measurement of the outcome, and selection of the reported result. The RoB in each domain was judged and classified into three levels: "low risk," "high risk," and "some concerns."

The methodological quality of evidence was systematically evaluated using the Grading of Recommendations, Assessment, Development and Evaluations approach.[22] Through this standardized framework, we assessed five domains that may downgrade evidence quality: (1) RoB (study limitations), (2) inconsistency (heterogeneity), (3) indirectness of evidence, (4) imprecision, and (5) publication bias. The evidence was categorized into four quality levels: high, moderate, low, and very low.

Given that many studies reported multiple effect sizes, we employed a three-level random-effects model to more accurately account for the correlations among effect sizes. The model parameters were initially estimated using the restricted maximum likelihood method to ensure robustness. For continuous outcomes, we utilized the standardized mean difference as the effect size measure. Considering the relatively small sample sizes of most included studies, we adopted Hedge's g, which is based on an exact formula and adjusted for small sample bias, as the effect size indicator. The magnitudes of effect sizes were categorized as follows: 0–0.2 for trivial, 0.2–0.5 for small, 0.5–0.8 for medium, and >0.8 for large. Heterogeneity was assessed using tau-squared (τ^2) and I-squared (I^2) statistics. The interpretation of I^2 was as follows: 0%–25% indicated low heterogeneity, 25%–75% indicated moderate heterogeneity, and >75% indicated high heterogeneity.

To evaluate the robustness of the results, sensitivity analyses were conducted, including leave-one-out analysis and changes in model structure, such as traditional two-level model and robust variance estimation (RVE). Additionally, subgroup analyses were performed based on different subcomponents of motor skills (gross motor skills, e.g., locomotion, objective control, body coordination, balance, strength, flexibility; and fine motor skills, e.g., manual dexterity, hand-eye coordination, handwriting), as well as control group type (passive control, active control), age group (younger age group: 5–8 years; older age group: ≥9 years), intervention frequency (<3 sessions/week, ≥3 sessions/week), duration (≤45 min, >45 min), period (≤8 weeks, >8weeks), and total volume(≤720 min, >720 min). Since most studies did not report intervention intensity, no analysis was conducted on this variable. The total intervention volume (in minutes) was defined as session duration × frequency × period. Publication bias and small-study effects were detected using funnel plots and Egger's regression test. If publication bias was detected, the trim-and-fill method was used for adjustment.

To model the potential nonlinear dose–response relationship, we employed restricted cubic splines (RCS) rather than standard linear meta-regression, as preliminary visual inspection of the data suggested a nonlinear trend (e.g., diminishing returns at higher volumes) and RCS can capture complex dose–response patterns without a priori specification of functional forms. The relationship between the total intervention volume (defined as session duration × frequency × period, in minutes), a pragmatic indicator of cumulative exposure time used due to the frequent lack of reported intensity data, and effect size was calculated. The RCS model was specified with 3 knots, which is consistent

with standard practice in meta-analytic dose–response modeling for small-to-moderate sample sizes. The knots were placed at the 10th, 50th, and 90th percentiles of the total exercise volume distribution, a strategy that balances flexibility (to capture nonlinear relationships) and stability (to avoid overfitting with limited data points). The RCS model was embedded within the three-level random-effects framework (used for the primary meta-analysis), which accounts for two levels of heterogeneity: within-study variance (between effect sizes from the same study) and between-study variance. Importantly, these dose–response findings, including the approximate thresholds, must be considered exploratory, given the inherent limitations of pooled analyses across heterogeneous exercise modalities. All statistical analyses were conducted using R software (version 4.2.1).

Results

A comprehensive search was conducted across electronic databases, yielding 1365 records. An additional six records were identified through manual reference screening. After removing duplicates, 962 unique records remained for title and abstract screening. Two independent reviewers performed the screening process, demonstrating good agreement ($\kappa = 0.70$, 95% CI: 0.60–0.82). Of these, 34 full-text articles were assessed for eligibility. Thirteen studies were excluded for the following reasons: (1) involving non-ADHD populations ($n = 2$), (2) reporting irrelevant outcomes ($n = 3$), (3) providing insufficient data ($n = 3$), (4) employing nonclinical study designs ($n = 2$), or (5) having inaccessible full texts ($n = 3$). Inter-rater agreement during full-text review was excellent ($\kappa = 0.88$, 95% CI: 0.72–0.99). Ultimately, 21 studies met the predefined eligibility criteria and were included in both qualitative synthesis and quantitative meta-analysis. The study selection process adhered to PRISMA guidelines, with details provided in Figure 1 and Supplementary Material Table S2–S3.

This systematic review and meta-analysis incorporated 21 studies (11 randomized controlled trials and 10 quasi-experimental studies) published between 2012 and 2024. A total of 850 participants were included across the studies, predominantly male (85.8%), with ages spanning 5–14 years. The studies were conducted primarily from Asia ($n = 11$), followed by Europe ($n = 7$), and North America ($n = 3$). Notably, only two studies specifically recruited children with ADHD without taking medicines. Of those studies, the most common comparator was a passive control with no intervention or waiting list ($n = 15$), followed by active control with routine school activities ($n = 5$). The physical exercise interventions spanned 3–24 weeks (median: 8 weeks), with session durations ranging from 15–90 min and frequencies of 1–6 sessions/week. Thirteen studies did not report exercise intensity, while the remaining eight studies employed moderate-to-vigorous intensity. Motor skill assessments were conducted using standardized tools, with gross motor skills evaluated in 18 studies and fine motor skills in 14 studies. The predominant assessment instruments were: Bruininks–Oseretsky Test of Motor Proficiency, 2nd Edition, Test of Gross Motor Development-2/3, and Movement Assessment Battery for Children-2. Detailed study characteristics are provided in Table 1.

The RoB assessment for included studies is summarized in Figure 2. Overall, 38.1% of studies demonstrated a low RoB, while 57.1% were rated as high risk, and 4.8% raised some concerns. Ten studies (47.6%) failed to adequately describe randomization procedures, introducing potential selection bias. One study (4.8%) was flagged for high risk due to excessive participant attrition (>20%), compromising data completeness. All studies satisfied criteria for outcome measurement and selective reporting, indicating consistent endpoint assessment and transparency in result presentation.

Effects on gross motor skills

The meta-analysis examining the effects of exercises on gross motor skills in children with ADHD included data from 16 studies, comprising 46 effect sizes and a total of 613 participants. The results indicated a moderate and positive effect of exercises on improving gross motor skills in children with ADHD (Hedge's $g = 0.62$, 95% CI: 0.48–0.76, $P < 0.001$, $I^2 = 32.61\%$). To assess the robustness of the results, we conducted a sensitivity analysis by

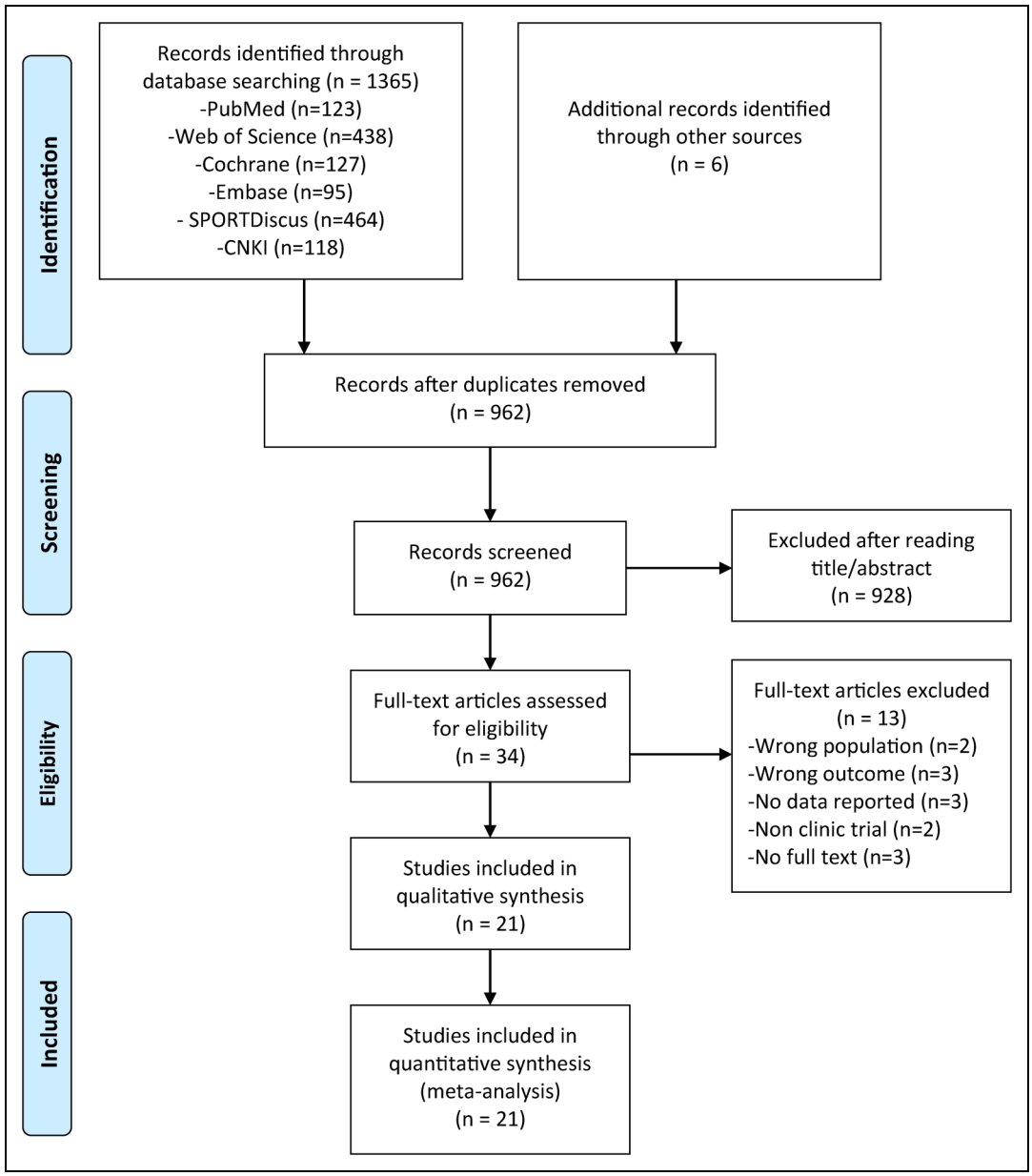


Figure 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram of the search process for studies.

sequentially excluding individual studies, which the effect size Hedge's g ranged from 0.59 to 0.64. Additionally, we compared the results using a traditional two-level model and a RVE method, which the traditional two-level model yielded a pooled effect size of Hedge's $g = 0.60$ (95% CI: 0.48–0.73), while the RVE method resulted in Hedge's $g = 0.62$ (95% CI: 0.48–0.76). The detailed results are presented in Figure 3A and Supplementary Material Figure S1.

Table 1. Characteristics of studies included in the review.

Study	Country/ region	Study design	N	Sex (M/F)	Age (year) mean (SD) or range	Physical exercises					Control arm	Outcomes	Medication use
						Type	Intensity	Duration (minutes)	Frequency (sessions/ week)	Week			
Benzing et al. (2019)[23]	Switzerland	RCT	51	43/8	10.63 (1.32)	Exergaming	MVPA	30	3	8	Waiting-list	Gross motor (German Motor Test)	Yes
Chang et al. (2022)[24]	Taiwan	RCT	48	39/9	8.36 (1.27)	Table tennis training/ Exergame table tennis training	NA	60	3	12	No intervention	Fine motor (Computerized handwriting movement assessment)	Yes
Chang et al. (2014)[25]	Taiwan	Quasi-experiment	27	23/4	8.44 (8.29)	Aquatic exercise	MPA	90	2	8	Waiting-list	Gross and fine motor (Basic Motor Ability Test-Revised)	Yes
Golubovic et al. (2014)[26]	Serbia	Quasi-experiment	36	36/0	6.38 (1.0)	Comprehensive physical fitness training	NA	60	3	24	No intervention	Gross motor (The battery of the six administered motor tests)	NA
Guo et al. (2023)[27]	China	RCT	40	36/4	7.60 (0.90)	Comprehensive physical fitness training	MVPA	90	6	4	No intervention	Gross motor (TGMD-2)	Yes
Jeyanthi et al. (2021)[28]	India	Quasi-experiment	20	20/0	8–12	Structured exercise program which included aerobics, resistance exercises, motor skills and attention training	NA	45	3	6	No intervention	Fine motor (Nine hole peg test)	NA
Kosari et al. (2012)[29]	Iran	Quasi-experiment	20	20/0	8.80 (0.70)	SPARK motor skill training	NA	45	3	6	No intervention	Gross motor (BOT-2)	Yes
Li et al. (2024)[30]	China	Quasi-experiment	32	NA	7.93 (1.61)	Ball physical activity program	MVPA	60	2	8	Routine school activities	Gross motor (TGMD-3)	NA
Lian et al. (2022)[31]	China	RCT	40	40/0	8.97 (1.91)	Motor skill training	MPA	35	3	12	Routine school activities	Gross and fine motor (BOT-2)	NA
Messler et al. (2016)[32]	Germany	RCT	28	28/0	11.00 (1.00)	HIIT	VPA	30	3	3	Group activity	Gross and fine motor (MABC-2)	Yes
Orangi et al. (2021)[33]	Iran	Quasi-experiment	36	36/0	10–13	Aerobic exercise	NA	60	3	8	Routine school activities	Gross and fine motor (BOT-2)	NA

(continued)

Table 1. Continued.

Study	Country/ region	Study design	N	Sex (M/F)	Age (year) mean (SD) or range	Physical exercises					Control arm	Outcomes	Medication use
						Type	Intensity	Duration (minutes)	Frequency (sessions/ week)	Week			
O'Connor et al. (2014)[34]	USA	Quasi-experiment	98	73/25	5–8	Summer Treatment Program	NA	60	5	8	Routine school activities	Gross and fine motor (BOT-2)	Yes
Pan et al. (2014)[12]	Taiwan	Quasi-experiment	24	24/0	7–14	SDHRP	NA	90	1	12	No intervention	Gross and fine motor (BOT-2)	Yes
Pan et al. (2016)[35]	Taiwan	RCT	32	32/0	6–12	Table tennis exercise	NA	70	2	12	No intervention	Gross and fine motor (BOT-2)	Yes
Pan et al. (2015)[14]	Taiwan	Quasi-experiment	30	30/0	7–12	Table tennis program	NA	70	2	12	No intervention	Gross motor (TGMD-2)	Yes
Silva et al. (2019)[13]	Brazil	RCT	20	14/6	11–14	Swimming training	NA	45	2	8	No intervention	Gross motor (The battery of corporal coordination tests for children)	Yes
Taft et al. (2015)[36]	Iran	RCT	24	20/4	8.10 (1.80)	Perceptual-motor training	NA	NA	3	6	Drug therapy	Gross and fine motor (BOT-2)	No
Torabi et al. (2018)[37]	Iran	Quasi-experiment	50	20/30	12.70 (1.08)	HIIT	VPA	15	3	6	No intervention	Gross and fine motor (BOT-2)	NA
Weerdmeester et al. (2016)[38]	Netherlands	RCT	73	58/15	9.77 (1.74)	Exergaming	NA	15	2	3	Routine school activities	Gross and fine motor (MABC-2)	Yes
Zhu et al. (2024)[39]	China	RCT	78	78/0	9.65 (1.06)	Gross and fine motor skill training	MPA	60	3	8	Waiting-list	Fine motor (MABC-2, Evaluation Tools of Children's Handwriting, and Chinese Handwriting Assessment)	Yes
Ziereis et al. (2015)[40]	Germany	RCT	43	32/11	9.45 (1.43)	Motor skill training/ Sports activities	NA	60	1	12	No intervention	Gross and fine motor (MABC-2)	No

BOT-2: Bruininks–Oseretsky Test of Motor Proficiency, 2nd Edition; HIIT: high-intensity interval training; MABC-2: Movement Assessment Battery for Children-2; MPA: moderate intensity physical activity; MVPA: moderate-to-vigorous intensity physical activity; NA: not reported; RCT: randomized controlled trial; SDHRP: simulated developmental horse-riding program; TGMD-2/3, Test of Gross Motor Development-2/3; VPA: vigorous intensity physical activity.



Figure 2. Risk of bias assessment for the included studies.

The meta-analysis revealed differential effects of exercise interventions across specific subcomponents of gross motor skills in children with ADHD. Results revealed exercise significantly improved locomotion skills (Hedge's $g = 0.53$, 95% CI: 0.19–0.87, $P = 0.002$, $I^2 = 0\%$), objective control (Hedge's $g = 1.20$, 95% CI: 0.48–1.92, $P = 0.001$, $I^2 = 64.19\%$), body coordination (Hedge's $g = 0.58$, 95% CI: 0.21–0.96, $P = 0.002$, $I^2 = 20.33\%$), and strength (Hedge's $g = 0.51$, 95% CI: 0.24–0.77, $P < 0.001$, $I^2 = 0.01\%$). Effects on balance (Hedge's $g = 0.37$, 95% CI: −0.10 to 0.84, $P = 0.124$, $I^2 = 0\%$) and flexibility (Hedge's $g = 0.50$, 95% CI: −0.16 to 1.16, $P = 0.139$, $I^2 = 62.25\%$) were not significant.

The detailed results are shown in Figure 4A–F and Supplementary Material Figures S2–S7.

Effects on fine motor skills

The meta-analysis investigating the impact of exercises on fine motor skills in children with ADHD encompassed 13 studies, 45 effect sizes, and 553 participants in total. It revealed a moderate positive effect (Hedge's $g = 0.72$, 95% CI: 0.54–0.90, $P < 0.001$, $I^2 = 49.71\%$), suggesting that exercises significantly enhanced fine motor skills in children with ADHD. To evaluate the stability of the results, a sensitivity analysis was conducted by sequentially excluding each study, which revealed that the Hedge's g varied between 0.69 and 0.76. Furthermore, the robustness of the results was verified through comparison with both a traditional two-level model (Hedge's $g = 0.78$, 95% CI: 0.64–0.92) and a RVE method (Hedge's $g = 0.72$, 95% CI: 0.54–0.90). The results are shown in Figure 3B and Supplementary Material Figure S8.

The meta-analysis revealed differential effects of exercise interventions across specific subcomponents of fine motor skills in children with ADHD. Results demonstrated that exercises significantly improved manual dexterity (Hedge's $g = 0.66$, 95% CI: 0.08–1.25, $P = 0.026$, $I^2 = 73.34\%$) and hand-eye coordination (Hedge's $g = 0.57$, 95% CI: 0.25–0.88, $P < 0.001$, $I^2 = 24.34\%$). Additionally, handwriting skills showed a particularly significant improvement (Hedge's $g = 0.86$, 95% CI: 0.68–1.04, $P < 0.001$, $I^2 = 6.13\%$). The detailed results are presented in Figure 4G–1 and Supplementary Material Figures S9–S11.

Moderator analysis

The moderator analyses revealed significant differential effects across intervention characteristics. For gross motor skills, exercise demonstrated substantially greater benefits when compared to passive controls (Hedge's $g = 0.65$, 95% CI: 0.49–0.81) than active controls (Hedge's $g = 0.25$, 95% CI: −0.19 to 0.70). Notably, session duration emerged as a critical moderator ($P < 0.001$), with interventions >45 min showing nearly double the effect size (Hedge's $g = 0.75$, 95% CI: 0.60–0.90) of shorter sessions (Hedge's $g = 0.34$, 95% CI: 0.17–0.52). Frequency of ≥3 sessions/week yielded larger effects (Hedge's $g = 0.84$, 95% CI: 0.64–1.04) than less sessions (Hedge's $g = 0.49$, 95% CI: 0.29–0.62). The detailed results are presented in Table 2.

For fine motor skills, exercises demonstrated substantially greater benefits when compared to passive controls (Hedge's $g = 0.87$, 95% CI: 0.74–1.00) than active controls (Hedge's $g = 0.05$, 95% CI: −0.48 to 0.58) and session duration of >45 min yielded larger effects (Hedge's $g = 0.81$, 95% CI: 0.62–1.00) than less session duration (Hedge's $g = 0.40$, 95% CI: 0.04–0.75). The detailed results are presented in Table 2.

Publication bias

The funnel plot asymmetry and Egger's test ($P < 0.05$) for gross motor skills indicated the presence of publication bias and small-study effects. Trim-and-fill analysis adjusted the effect size downward to Hedge's $g = 0.44$ (95% CI: 0.29–0.59), suggesting an overestimation in the original analysis. Similarly, fine motor skills outcomes demonstrated funnel plot asymmetry with significant Egger's test results ($P < 0.05$), primarily reflecting small-study effects. The trim-and-fill adjusted effect size remained robust at Hedge's $g = 0.68$ (95% CI: 0.52–0.83), indicating relatively stable estimates despite potential bias. The detailed results are provided in Supplementary Material Figures S12–S13.

Quality of evidence

Pooled analyses yielded moderate-quality evidence for both gross motor skills (limited by RoB) and fine motor skills (limited by RoB). However, quality of evidence for specific subcomponents of motor skills ranged from very low to moderate (Supplementary Material Table S4).

Dose–response relationships

The effect size for gross motor skills showed a trend ($\beta = -0.01$, 95% CI: −0.43 to 0.40) of increasing with the total dose up to approximately 810 min,

Table 2. The subgroup meta-analysis of exercises on gross and fine motor skills.

Outcomes	Subgroup	*K*	Hedge's *g*	95% CI	*P*	I^2
Gross motor skills	Control group				0.049	
	Passive control	34	0.65	[0.49, 0.81]	<0.001	19.5%
	Active control	12	0.25	[−0.19, 0.70]	0.265	0%
	Age				0.596	
	Younger	13	0.68	[0.43, 0.93]	<0.001	15.4%
	Older	33	0.60	[0.42, 0.77]	<0.001	34.0%
	Frequency (sessions/week)		0.84	[0.64, 1.04]	0.004	
	<3	21	0.49	[0.29, 0.62]	<0.001	37.3%
	≥3	25	0.84	[0.64, 1.04]	<0.001	0%
	Session duration (minutes)				<0.001	
	≤45	18	0.34	[0.17, 0.52]	<0.001	0%
	>45	27	0.75	[0.60, 0.90]	<0.001	17.0%
	Length (weeks)				0.335	
	≤8	24	0.56	[0.37, 0.75]	<0.001	26.4%
	>8	22	0.70	[0.49, 0.91]	<0.001	28.8%
	Total dose (minutes)				0.102	
	≤720	17	0.45	[0.22, 0.67]	<0.001	16.8%
	>720	28	0.68	[0.51, 0.84]	<0.001	17.7%
Fine motor skills	Control group				0.003	
	Passive control	37	0.87	[0.74, 1.00]	<0.001	32.3%
	Active control	8	0.05	[−0.48, 0.58]	0.861	0%
	Age				0.258	
	Younger	12	0.51	[0.10, 0.91]	0.101	6.8%
	Older	33	0.77	[0.56, 0.98]	<0.001	50.9%
	Frequency (sessions/week)				0.554	
	<3	18	0.78	[0.49, 1.06]	<0.001	47.2%
	≥3	27	0.66	[0.41, 0.92]	<0.001	45.1%
	Session duration (minutes)				0.044	
	≤45	8	0.40	[0.04, 0.75]	0.030	76.6%
	>45	37	0.81	[0.62, 1.00]	<0.001	25.4%
	Length (weeks)				0.450	
	≤8	20	0.65	[0.39, 0.91]	<0.001	53.3%
	>8	25	0.79	[0.53, 1.05]	<0.001	38.3%
	Total dose (minutes)				0.834	
	≤720	11	0.73	[0.51, 0.95]	<0.001	68.0%
	>720	34	0.68	[0.33, 1.04]	<0.001	35.0%

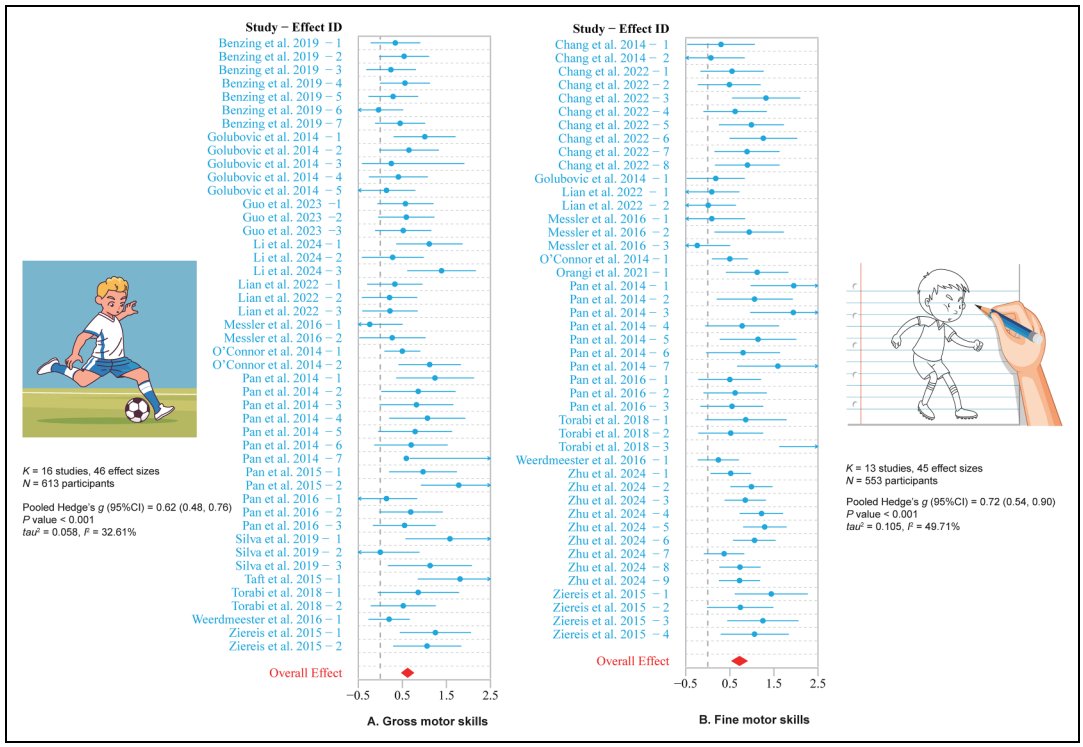


Figure 3. Forest plot for change in gross motor skills (A) and fine motor skills (B).

after which the curve plateaued (Figure 5A). For fine motor skills, a significant positive dose–response relationship was observed up to a cumulative dose of approximately 1350 min ($\beta = 0.18$, 95% CI: 0.03–0.34), beyond which further gains appeared to diminish (Figure 5B).

Discussion

This systematic review and meta-analysis comprehensively evaluated the effects of physical exercises on motor skills in children and adolescents with ADHD, focusing on both gross and fine motor skills and their subcomponents. The results demonstrated moderate positive effects on both gross (Hedge's $g = 0.62$) and fine motor skills (Hedge's $g = 0.72$), with significant improvements observed across specific subcomponents. Notably, exercise interventions exhibited differential efficacy based on skill type, intervention parameters, and control conditions, underscoring the importance of tailored approaches in clinical and educational settings.

The meta-analysis revealed that physical exercise interventions significantly improved gross motor skills in children with ADHD, with notable enhancements in locomotion, object control, body coordination, and strength. These improvements can be attributed to the structured and repetitive nature of exercise interventions, which provide opportunities for motor learning and skill acquisition.[30] In contrast, balance and flexibility did not show significant improvements. This may be due to the specific nature of the exercises included in the interventions. Balance requires precise control of the vestibular and proprioceptive systems, which may not be adequately targeted by the general exercise programs reviewed in this meta-analysis. Flexibility, on the other hand, is often improved through specific stretching and mobility exercises, which were not a primary focus of the interventions included in this study. The lack of significant effects on these

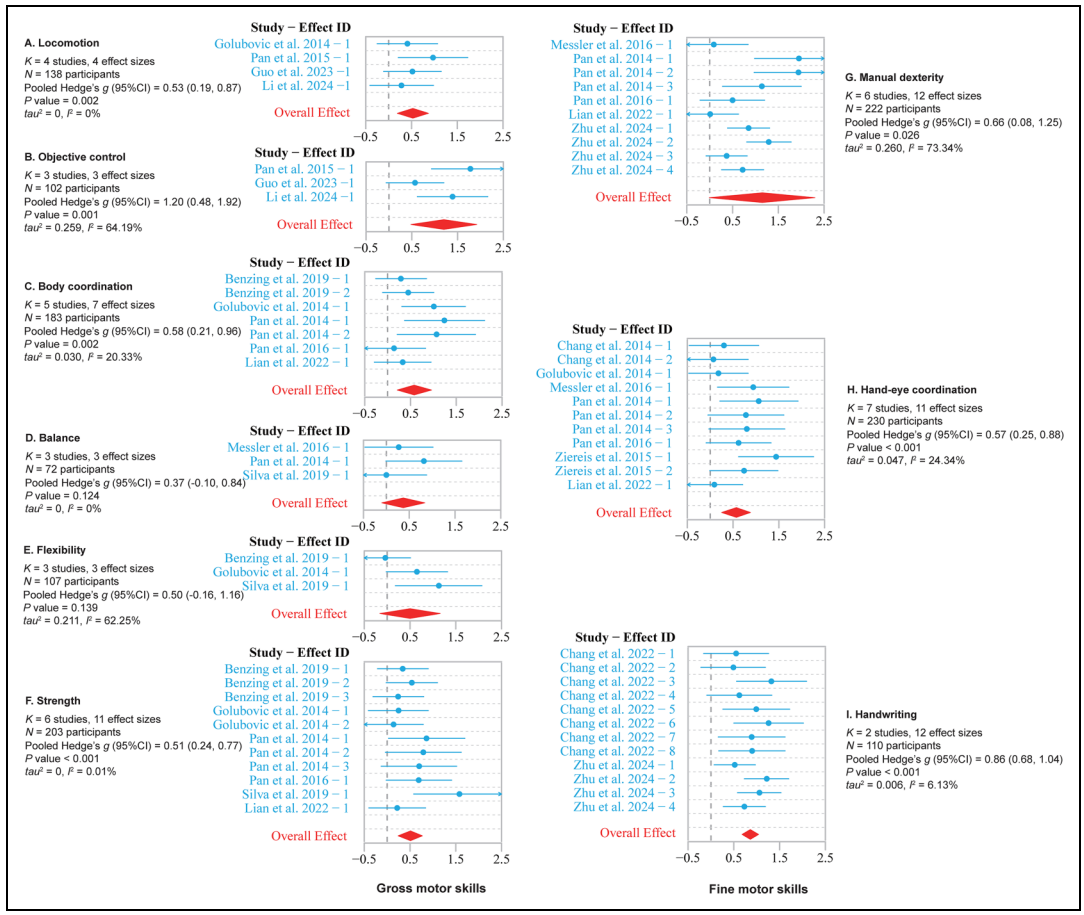


Figure 4. Forest plot for change in different subcomponents of gross motor skills (A–F) and fine motor skills (G–I).

subcomponents suggests that future interventions may need to incorporate more targeted exercises to address balance and flexibility deficits in children with ADHD. Additionally, the heterogeneity in the types of exercises and the varying durations and intensities of the interventions may have contributed to the inconsistent effects on balance and flexibility.

The observed improvements in fine motor skills, particularly handwriting, represented a critical finding. Unlike gross motor skills that primarily impact physical activities, fine motor deficits directly impair essential daily functions from self-care tasks like buttoning shirts to academic performance through handwriting difficulties. The robust effects likely stem from shared neural circuitry between fine motor control and the executive functions impaired in ADHD. Handwriting gains may reflect enhanced integration within the cerebello-thalamo-cortical circuit, which coordinates motor precision and cognitive control.[24,40] Exercise-induced neuroplasticity, including striatal and prefrontal dopamine upregulation, improved cerebellar connectivity, and parietal lobe adaptation could underpin these changes.[11] Similarly, gains in manual dexterity and hand-eye coordination suggested optimization of the dorsal visual stream pathway, crucial for translating visual input into precise movements. Importantly, these improvements may create a virtuous cycle: fine motor tasks inherently train executive functions through demands on working memory and inhibitory control, while enhanced cognitive functions in turn support better motor learning. This bidirectional

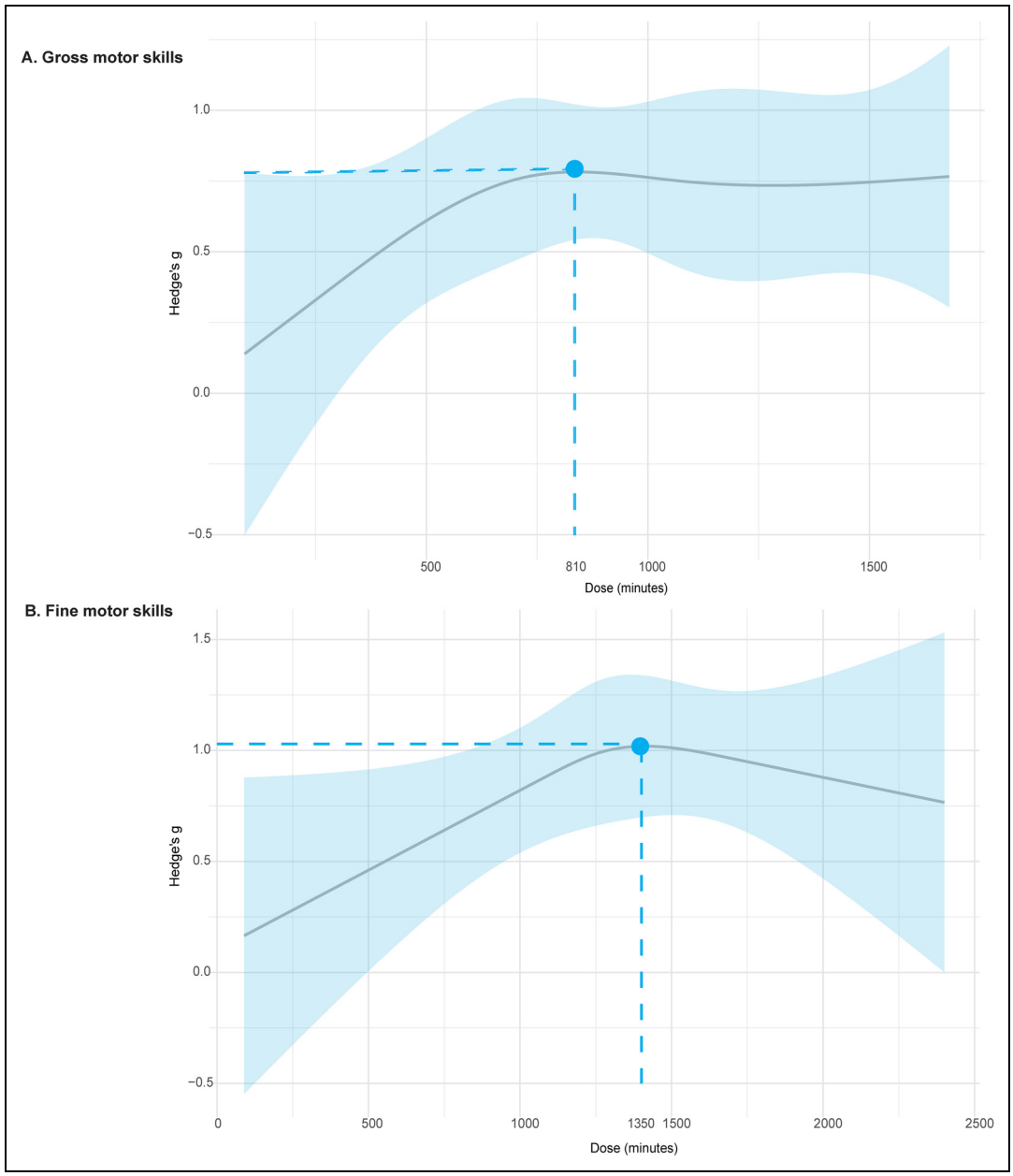


Figure 5. Dose–response relationships in gross motor skills (A) and fine motor skills (B).

relationship underscores why targeted fine motor interventions could address both the motor and core cognitive symptoms of ADHD, offering a comprehensive therapeutic approach that extends beyond conventional symptom management.

Moderator analyses indicated that higher intervention frequency (≥3 sessions/week) and longer session duration (>45 min) were associated with greater motor skill gains, and improvements in fine motor skills may require a longer period of accumulated

practice (~1350 min) than gross motor skills (~810 min) before plateauing. This discrepancy likely reflects distinct neurobiological demands. Fine motor control depends more heavily on the integration of prefrontal-executive and sensorimotor networks, which are particularly impaired in ADHD due to reduced dopamine availability and weaker fronto-cerebellar connectivity.[41] While gross motor skills may benefit from general physical activity through basic neuromuscular adaptations, fine motor tasks (e.g., handwriting, buttoning) demand precise coordination between cognitive control (e.g., working memory for letter formation) and hand musculature —a process requiring repeated, focused practice to induce cortical reorganization in the primary motor and somatosensory cortices.[42] Previous meta-analyses have similarly shown that executive functions (closely linked to fine motor control) require longer intervention durations to demonstrate improvement, suggesting that the extended dose requirement for fine motor gains may reflect their shared dependence on frontostriatal circuit plasticity.[43,44] These findings underscore the importance of sustained, frequent exercise protocols when targeting fine motor deficits in ADHD, as the neural systems supporting these skills appear to need greater cumulative stimulation for meaningful change. Therefore, tailored exercise programs that consider individual motor skill profiles and integrate specific motor training goals may be more effective in clinical and educational settings. We emphasized that the dose–response relationships are modality-agnostic and should be interpreted as "general cumulative exposure thresholds" rather than modality-specific prescriptions. For example, dynamic sports (e.g., table tennis, swimming) may achieve gross motor skill improvements at a lower volume due to their inherent focus on locomotion/object control, while precision-based exercises (e.g., perceptual-motor training) may require similar or higher volume for fine motor gains.

This study had several limitations. First, the high RoB in many included studies, often due to inadequate description of randomization, may affect the validity and precision of the pooled estimates. Second, inconsistent reporting of intervention details, particularly exercise intensity, which limited moderator analysis and the specificity of exercise prescriptions. Third, generalizability was constrained by the underrepresentation of participants not taking medications, limiting applicability to untreated populations. Fourth, evidence for specific motor subcomponents was of low or very low certainty due to imprecision and heterogeneity. Additionally, the dose–response analysis, based on cumulative exercise volume, did not account for unreported intensity or modality-specific demands, so the derived thresholds should be viewed as exploratory rather than prescriptive. Future studies should prioritize high-quality randomized controlled trials with standardized intensity monitoring, include medication-free cohorts, and further investigate effects on specific skill domains and underlying mechanisms.

In conclusion, this meta-analysis provided moderate evidence that physical exercises are an effective intervention for improving gross and fine motor skills in children and adolescents with ADHD. The benefits vary across different motor subcomponents, highlighting the importance of tailored exercise programs. Notably, fine motor skills appeared to require greater cumulative exercise exposure than gross motor skills to achieve substantial gains, suggesting that exercise intervention protocols should carefully consider dose parameters. These findings support the integration of structured physical activity into clinical and educational interventions for ADHD; however, these findings should be interpreted with caution due to the methodological limitations and confounding factors of the included studies.

Clinical messages

- Physical exercise programs should be designed to include longer (>45 min) and more frequent (≥3 sessions/week) sessions to maximize benefits on motor proficiency.
- Interventions should be specifically tailored to target different motor skill components.

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Author contributions

ZYX contributed to writing—original draft, methodology, investigation, formal analysis, and data curation. CXM contributed to writing—original draft, software, methodology, investigation, and formal analysis. DZH contributed to writing—original draft, validation, investigation, and data curation. SJX contributed to writing—review & editing, validation, resources, and formal analysis. TY contributed to writing—review & editing, investigation, and visualization. CB contributed to writing—review & editing, methodology, software, and data curation. ZFL contributed to writing—review & editing, conceptualization, supervision, and project administration.

Ethics approval and consent to participate

In this study, all methods were performed in accordance with the relevant guidelines and regulations. Ethics approval and consent to participate is not necessary since our study is a meta-analysis and systematic review.

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Data availability statement

The data supporting the conclusions of this review are included in this published article and in Supplemental Information Files.

Supplemental material

Supplemental material for this article is available online.

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